



## **Mini Project Report - 08**

Master of Computer Application – GENERATIVE AI  
Semester – I

**Sub: FRONT-END FRAMEWORKS AND TECHNOLOGIES**

**Topic: JAVASCRIPT GRADE CALCULATOR**

By

**Name: R KUSHMA**

**Reg no.: PROV/ASAC/MCA/7/26/031**

**Faculty Name: VEERA RAGHAV K**

**Faculty Signature: \_\_\_\_\_**

**Department of Computer Application  
Alliance University  
Chandapura - Anekal Main Road, Anekal  
Bengaluru - 562 106**

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# INTRODUCTION

The **Grade Calculator** is a simple web-based project developed using **HTML, CSS and JavaScript**. The main purpose of this project is to calculate and display a student's academic result based on the marks entered for different subjects.

The webpage provides a simple and user-friendly form where the student can enter their **name, roll number and marks** for subjects such as Maths, English, Python, Frontend Frameworks and Aptitude.

The **HTML** is used to create the structure of the webpage, including headings, labels, input fields, buttons and the result section. **CSS** is used to provide basic styling, colours, spacing and layout to make the calculator easy to use. **JavaScript** is used to perform the main calculations and provide dynamic results.

It calculates the **total marks, average, percentage, grade and pass/fail status** based on the entered marks. The project also includes validation to check whether valid details are entered and a **Reset** button to clear the form and results.

Overall, this project demonstrates the basic use of **HTML, CSS and JavaScript** to create an interactive web application.

# INPUT CODE

## Node app.js code:

```
let a = 10;
let b = 100;
console.log("sum of a,b:"+(a+b));

const fs = require("fs");
fs.writeFileSync("javascript Grade.html", "This is a javascript file created by node.js");
console.log("File Created...");

const os = require("os");
console.log(os.platform());

console.log(os.platform());
```

## sample file creation html code:

```
<html>
  <head>
    <title>JAVASCRIPT GRADE CALCULATOR</title>
  </head>
  <body>
    <h1>JAVASCRIPT GRADE CALCULATOR</h1>
  </body>
</html>
```

## Node Server.js code:

```
const http = require("http");
const fs = require("fs");

const server = http.createServer((req, res)=>
{
  fs.readFile("javascriptGradeCalculator.html", (err, data)=>{
    res.end(data);
  });
});

server.listen(2000);
console.log("http://localhost:2000");
```

## **HTML + CSS Code:**

```
<html>
<head>
  <title>Grade Calculator</title>

  <style>
    body {
      font-family: Arial;
      margin: 0;
      background-color: #e8e4e6;
    }

    header {
      background-color: #9c5b70;
      color: white;
      text-align: center;
      padding: 20px;
    }

    section {
      background-color: white;
      margin: 20px;
      padding: 20px;
    }

    form {
      display: flex;
      flex-direction: column;
      width: 300px;
    }

    input {
      padding: 8px;
      margin: 5px 0 10px;
    }

    input[type="submit"] {
      background-color: #333;
      color: white;
      border: none;
    }

    button {
      background-color: #333;
      color: white;
      padding: 8px;
      border: none;
    }

    .result {
      background-color: #eeeeee;
      padding: 10px;
      margin-top: 10px;
    }
  </style>
</head>
<body>
```

```

</style>
</head>

<body>

  <header>
    <h1><b>GRADE CALCULATOR</b></h1>
  </header>

  <section>

    <h2>STUDENT INFORMATION</h2>

    <form id="gradeForm">

      <label>NAME:</label>
      <input type="text" id="name">

      <label>ROLL NUMBER:</label>
      <input type="text" id="roll">

      <label>MATHS:</label>
      <input type="number" id="maths">

      <label>ENGLISH:</label>
      <input type="number" id="english">

      <label>PYTHON:</label>
      <input type="number" id="python">

      <label>FRONTENDED FRAMEWORKS:</label>
      <input type="number" id="frontened frameworks">

      <label>APTITUDE:</label>
      <input type="number" id="aptitude">

      <input type="submit" value="Calculate">

    </form>

    <button onclick="resetForm()">Reset</button>

    <div class="result">
      <h2>Result:</h2>

      <p id="total"></p>
      <p id="average"></p>
      <p id="percentage"></p>
      <p id="grade"></p>
      <p id="status"></p>

    </div>

  </section>

```

```

<script>

document.getElementById("gradeForm").onsubmit = function(event) {

    event.preventDefault();

    var NAME = document.getElementById("name").value;
    var MATHS = Number(document.getElementById("maths").value);
    var ENGLISH = Number(document.getElementById("english").value);
    var PYTHON = Number(document.getElementById("python").value);
    var FRONTENED_FRAMEWORKS = Number(document.getElementById("frontened
frameworks").value);
    var APTITUDE = Number(document.getElementById("aptitude").value);

    if (NAME == "" || MATHS < 0 || ENGLISH < 0 || PYTHON < 0 ||
FRONTENED_FRAMEWORKS < 0 || APTITUDE < 0) {
        alert("Please enter valid details");
        return;
    }

    var total = MATHS + ENGLISH + PYTHON + FRONTENED_FRAMEWORKS +
APTITUDE;
    var average = total / 5;
    var percentage = total / 5;
    var grade;

    if (percentage >= 90) {
        grade = "A+";
    }
    else if (percentage >= 80) {
        grade = "A";
    }
    else if (percentage >= 70) {
        grade = "B";
    }
    else if (percentage >= 60) {
        grade = "C";
    }
    else if (percentage >= 50) {
        grade = "D";
    }
    else {
        grade = "F";
    }

    var status;

    if (percentage >= 40) {
        status = "Pass";
    }
    else {
        status = "Fail";
    }

    document.getElementById("total").innerHTML = "Total Marks: " + total;

```

```
document.getElementById("average").innerHTML = "Average: " + average;
document.getElementById("percentage").innerHTML = "Percentage: " + percentage + "%";
document.getElementById("grade").innerHTML = "Grade: " + grade;
document.getElementById("status").innerHTML = "Result: " + status;

};

function resetForm() {

    document.getElementById("gradeForm").reset();

    document.getElementById("total").innerHTML = "";
    document.getElementById("average").innerHTML = "";
    document.getElementById("percentage").innerHTML = "";
    document.getElementById("grade").innerHTML = "";
    document.getElementById("status").innerHTML = "";
}

</script>

</body>
</html>
```

# OUTPUT CODE

The screenshot shows a web browser window with the address bar displaying 'localhost:2000'. The page title is 'GRADE CALCULATOR'. The main content area is titled 'STUDENT INFORMATION' and contains several input fields with the following values:

- NAME: KUSHMA
- ROLL NUMBER: 22
- MATHS: 67
- ENGLISH: 88
- PYTHON: 78
- FRONTEND FRAMEWORKS: 80
- APTITUDE: 100

Below the input fields is a 'Calculate' button. A 'Reset' button is also visible. The 'Result' section displays the following calculated values:

- Total Marks: 411
- Average: 82.2
- Percentage: 82.2%
- Grade: A
- Result: Pass

# CONCLUSION

The Grade Calculator project successfully provides a simple and efficient way to calculate a student's academic performance. The project combines HTML for webpage structure, CSS for styling and JavaScript for calculations and interaction.

Based on the marks entered by the user, the program calculates the total, average and percentage and assigns a grade from A+ to F. It also determines whether the student has passed or failed based on the percentage.

The use of JavaScript conditions such as if, else if and else helps in determining the appropriate grade and result status. The form validation ensures that incorrect or negative marks are not accepted. The Reset function allows the user to clear all the entered information and calculated results.

This project helped in understanding important concepts such as HTML forms, CSS styling, JavaScript variables, functions, event handling, input validation, arithmetic calculations and conditional statements. It provides a basic understanding of how HTML, CSS and JavaScript work together to create an interactive webpage.